

MATTHEW AZER

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EDUCATION

Rutgers University - School of Engineering

Bachelor of Science in Electrical & Computer Engineering

New Brunswick, NJ

Expected May 2028

Relevant Coursework

Integrated Data-Driven Design for Engineering Applications, Engineering Mechanics: Statics, Principles of Electrical Engineering, Digital Logic Design, Probability & Random Processes, Programming Methodology, Computer Architecture, Linear Systems & Signals

PROJECTS

Autonomous Word Hunt Robot

- Developed a Python-based computer vision system using OpenCV and Tesseract OCR to recognize a live 4x4 letter grid and identify high-scoring words from a database of 129,000+ candidate words using trie-based depth-first search and path-planning algorithms.
- Programmed a GRBL-controlled XY plotter with a servo-actuated capacitive stylus to autonomously trace and submit 45+ words per game, optimizing motion speed, acceleration, and touchscreen interaction for reliable operation.

Four-Way Traffic Light Controller

- Designed and built a four-way traffic light control circuit using AND/OR/NOT gates, JK/D flip-flops, counters, and a 555 timer to automatically sequence 12 red, yellow, and green LEDs across an intersection.
- Implemented and tested sequential and combinational logic on a breadboard, troubleshooting circuit timing, wiring, and logic errors to achieve reliable operation across 100+ continuous signal cycles.

WORK EXPERIENCE

MasTec Utility Services (PSE&G Contract)

Field Technician Intern

New Brunswick, NJ

May 2026 - Aug 2026

- Performed cathodic protection testing across 500+ test stations, collecting and analyzing 500+ voltage measurements to assess corrosion protection and identify 25+ abnormal conditions on PSE&G underground gas pipeline systems.
- Utilized digital reporting software to document and validate 500+ field measurements, verify measurement accuracy, and assist in troubleshooting abnormal system readings, reducing repeat field measurements by approximately 15%.

St. Mary Summer Camp

Head Counselor

East Brunswick, NJ

June 2022 - Aug 2025

- Supervised and mentored students, leading to improved academic performance and engagement in activities.
- Delivered mentorship and ensured structured communication with parents and staff.

LEADERSHIP & INVOLVEMENT

N2E Robotics Club (Rutgers University)

Member

New Brunswick, NJ

Sep 2024 - Present

- Participate in hands-on robotics projects involving ROS, robot simulation, sensors, and programming, applying engineering concepts to build and test robotic systems.
- Collaborate with other students during technical workshops to troubleshoot hardware and software, configure robotic systems, and develop solutions for autonomous and simulated robot applications.

St. Mark Homeless Service

Service Leader

New Brunswick, NJ

Sep 2024 - Present

- Lead and coordinate a campus-based outreach initiative providing food, clothing, and essential supplies directly to individuals experiencing homelessness throughout New Brunswick.
- Organize volunteers, manage donation collection and preparation, and lead local distribution efforts to expand the service's reach within the surrounding community.

SKILLS

Languages: Python, C, C++, Java, MATLAB, Verilog, Assembly, RISC-V Assembly, G-code

Hardware & EDA Tools: LTSpice, Xilinx Vivado, Arduino IDE, Raspberry Pi, AutoCAD, SolidWorks, Onshape, IBM Rhapsody, Soldering, Circuit Testing, Digital Multimeter, Frequency Measurement, Serial Communication

Software & Libraries: OpenCV, Tesseract OCR, Linux, Git/GitHub, VS Code, Cursor, ROS